Journal of Physics:
Conference Series

PURPOSE-LED PUBLISHING™

PAPER • OPEN ACCESS

Development of the front-end board of a Xenon gas Time Projection Chamber at the AXEL neutrinoless double beta decay search experiment

To cite this article: Shunsuke Tanaka and for the AXEL collaboration 2017 *J. Phys.: Conf. Ser.* **888** 012075

View the article online for updates and enhancements.

You may also like

- Multiple-isotope comparison for determining 0 mechanisms
V M Gehman and S R Elliott
- Double beta decay experiments
K Zuber
- Neutrinoless double-beta decay and neutrino physics
Werner Rodejohann

Enabling Electrochemistry Inside and Outside the Lab

PalmSens

Discover our diverse range of Potentiostats engineered to support your research wherever it takes you

- Intuitive software for Windows and Android
- Proven in over 4000 scientific publications
- Single-channel and multi-channel
- Benchtop, portable and wearable

Discover More



This content was downloaded from IP address 192.51.222.137 on 30/10/2026 at 07:59

IOP Conf. Series: Journal of Physics: Conf. Series **888** (2017) 012075 doi:10.1088/1742-6596/888/1/012075

Development of the front-end board of a Xenon gas Time Projection Chamber at the AXEL neutrinoless double beta decay search experiment

Shunsuke Tanaka

for the AXEL collaboration

Kyoto University, Kitashirakawa Oiwake-cho Sakyo-ku Kyoto-shi Kyoto-fu, Japan

E-mail: syun.tanaka@scphys.kyoto-u.ac.jp

Abstract. AXEL is a project to search for 0νββ using a High pressure Xenon gas TPC. AXEL uses SiPM's to measure the energies and the tracks of 0νββ events. About 50,000 SiPM's are required for final 0νββ searching version, so developing Front-End Boards (FEB) are necessary. We develop FEB that has high energy resolution and wide dynamic range.

1. Introduction

AXEL (A Xe ElectroLuminescence detector) is a project to search for neutrinoless double beta decay (0νββ) of ^{136}Xe, using a High pressure Xenon gas Time Projection Chamber (TPC). Figure 1 is the schematics of the final ton-scale version of AXEL detector. AXEL has following 3 features: High energy resolution (aim at 0.5% FWHM @2.4MeV, Q-value of 0νββ of ^{136}Xe), Tracking for background rejection, and Extendable to large size.

These features are based on ELCC (Electroluminescence Light Collection Cell) plane, that we are developing own (Figure 2). Ionized electrons drift along the electric field and they are collected into cells of ELCC plane. These electrons are accelerated and emit photons by the EL (ElectroLuminescence) process. EL process is the intermittent excitation-deexcitation of atoms (Figure 3). There is good linearity between the number of the emitted photons and the collected electrons, so energy of event can be measured correctly. Event topology is able to be reconstructed from the position of cells and the timing of the signals. Additionally ELCC contain solid PTFE plate, so it can be easily extended.

ELCC has many SiPM's to detect EL photons. The number of SiPM's is 50,000 in the final ton-scale version, and in the next kg-scale prototype detector what starts measurement by the next year at the earliest, even contains 1,000 SiPM's. We start developing the Front-End Board (FEB).

2. Basic structure of FEB

The requirements to FEB are following : High channel density for low cost, Recording waveform for tracking, Adjustable system of high voltage for each SiPM, High resolution for charge of event, and Large dynamic range for pulse width (5 - 300 us) and number of photons (1 p.e. - 10^5 p.e.).

Content from this work may be used under the terms of the Creative Commons Attribution 3.0 licence. Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.
Published under licence by IOP Publishing Ltd

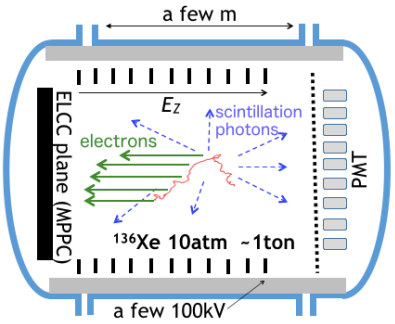


Figure1. Schematic of AXEL

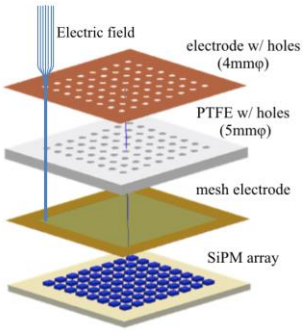


Figure2. ELCC plane

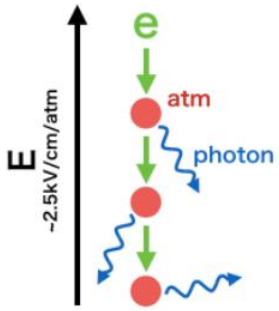


Figure3. EL process

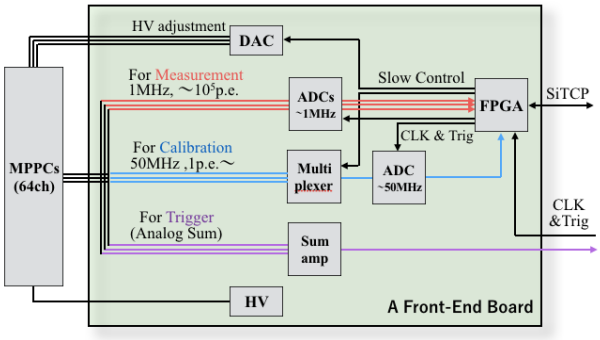


Figure4. Construction of FEB

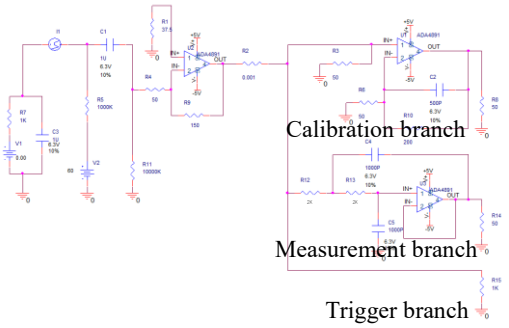


Figure5. Design of Analog process part

To satisfy these requirements, we designed the FEB as Figure 4.

There are DAC on the FEB to adjust the high voltage for each SiPM's. Common HV source apply same voltage to all SiPM's and then DAC adjust the ground level depending on each SiPM. Thus the gain of each SiPM is able to be equal. Additionally to get wide dynamic range, we use 2 kinds of ADCs. One is the 50 MSPS for taking dark (1p.e.) signal, and the other is 1MSPS for measuring the energy of event. The signals of SiPM's are divided to 3 ways, one of them is summed up with other signals and used for trigger, and other 2 signals are digitized and measured by these ADCs.

Figure 5 is the analog process part to divide signal. Calibration branch is high gain (×15), smoothed by 100ns filter and sampled by 50MSPS. On the other hand, Measurement branch is low gain (×3), smoothed by 2us filter and sampled by 1MSPS.

3. Evaluation of the analog process part

The Analog part is evaluated by 2 ways, simulation and the prototype circuit.

3.1. Simulation

There are 2 kinds of important signals for this experiment. One is the 0νββ signal, the energy is 2458keV and ELCC plane collect about 10^5 electrons in 10us - 300us. The other is a characteristic X-ray from Xenon that is used for energy calibration. The energy of X-ray is 30keV and about 1200 electrons are collected in 4 - 8us.

To simulate these signals, we use some software's. Geant4 generate 30keV X-ray or 2 electrons from 0νββ and calculate the number and the position of electrons ionized by them. On the other hand, Garfield++ accelerate an electron and calculate the number of the EL photons emitted in ELCC plane. Combine these simulations, we get the number and the timing of photons entered into SiPM's. Then, SPICE simulate the response of the circuit against the input signal from SiPM's. Output signals are digitized and sampled by 1 or 50 MSPS. We evaluated the fluctuation of the sum of output signals.

Table1. Simulation results

fluctuation of	Number of photons	Integral of Calibration	Integral of Measurement	Contribution of Calibration branch	Contribution of Measurement branch
30keV_12bit [%(FWHM)]	2.74	2.73	2.85	～0	0.78
2458keV_12bit [%(FWHM)]	0.67	—	0.67	—	0.53
2458keV_14bit [%(FWHM)]	0.41	—	0.46	—	0.21

The results are listed on Table1. The response of the 30keV signal reach good resolution, compared with the aim (0.5% FWHM @2.4MeV). Although the response against 2458keV signal what digitized by 12bit don't have enough resolution, the resolution reaches 0.46% FWHM at the 14bit digitizing one. The contribution of circuit of the value is only 0.21%.

3.2. Prototype circuit

We also make prototype circuit. Function Generator input the rectangle signals into the circuit, and digitize both the input and the output signals by digitizer (CAEN V1724: 100MSPS, 14bit). These data are down sampled to 1MSPS and integrated. Then we evaluated the fluctuation of the ratio of them.

The results are listed on Table2. We inputted same charge signals (pulse height × pulse width = const. and the charge is the same value of 0νββ signal). The pulse height is scanned between 0.20V - 1.00V and the pulse width is changed between 50 - 250us depending on the pulse height. First, we measure the fluctuation against 1,000 same shape pulses. Though the fluctuation is changed by the shape of pulse, the fluctuation value is between 0.03 - 0.1% FWHM. Second, we measure the fluctuation among different pulses. The value of fluctuation is 0.039% (r.m.s.) and the convert value is 0.092% (FWHM). They are also good value comparing the aim value.

Table2. Results of prototype circuit

	Same shape	Same charge
Fluctuation of Input/Output [% (FWHM)]	0.03 ～ 0.1 (depends on the pulse shape)	0.092 (converted by std.dev × 2.35)

4. Summary

AXEL is the project to search 0νββ using ^{136}Xe. We are developing the FEB for AXEL that has high channel density, wide dynamic range, and high resolution for charge of event.

We evaluated the analog process part of the FEB. Simulation shows the resolution of the circuit against the 0νββ signal is 0.46% FWHM. Additionally, the prototype circuit shows the fluctuation against the same rectangle pulses is less than 0.1% FWHM, and that against the same charge pulse is 0.092% FWHM (converted).

Towards the start of the measurement using the 1,000 sh next prototype detector, we will have completed this FEB and evaluates them soon.

Acknowledgements

This work was partially supported by KAKENHI Grant-in-Aids for Scientific Research (A) (JP15H02088) and JSPS KAKENHI Grant Number 16J09462.